IS 4108- IS Strategies

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LECTURE 09- MANAGING INFORMATION SYSTEMS INVESTMENTS

What is an IS investment?

- An Information Systems investment refers to the allocation of organizational resources to acquire, develop, implement, maintain, or upgrade information systems and technologies.

What is an IS investment?

- Information systems (IS) investments are strategic capital expenditures into hardware, software, databases, networks, and personnel.
- Organizations fund these systems to optimize operations, enhance data-driven decision-making, and gain a sustainable competitive edge.

Examples

Investments on;

- Enterprise Resource Planning (ERP) systems
- Customer Relationship Management (CRM) systems
- Cloud computing solutions
- Cybersecurity infrastructure
- Artificial Intelligence applications

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INVESTMENTS IN AN INFORMATION SYSTEM
The five key areas organizations invest in to build, implement and run successful systems
1. PEOPLE
(Users and IT Staff)
Hiring specialized talent
Funding change management programs
Training employees to use the new system
Examples
• Systems analysts
• Project managers
• Change management teams
• Training programs
2. PROCESSES
(Workflows and Procedures)
Hiring consultants to redesign business operations
Document new rules and procedures
Align daily habits with the new technology
Examples
• Process redesign
• Workflow documentation
• Standard operating procedures
• Policy updates
3. DATA
(Information Assets)
Funding data migration
Cleaning legacy databases
Establishing data governance and security protocols
Examples
• Data migration tools
• Data quality programs
• Data governance frameworks
• Security and compliance
4. SOFTWARE
(Instructions)
SaaS
SOFTWARE LICENSE
Buying software licenses
Paying for SaaS subscriptions
Hiring developers for custom coding
Examples
• Application licenses
• SaaS subscriptions
• Development tools
• Custom software
5. HARDWARE
(Physical Infrastructure)
Purchasing physical servers, computers
Networking cables and equipment
Storage drives and mobile devices for workers
Examples
• Servers
• Computers and laptops
• Networking equipment
• Storage devices
• Mobile devices

Importance of Managing IS Investments

IS investments often require significant financial and human resources.

Challenges

- High implementation costs
- Technology obsolescence
- Project failures
- Security risks
- Uncertain returns

Therefore, organizations must carefully evaluate and manage these investments.

Categories of IS Investments

Infrastructure Investments
Support all organizational operations.

Ex:
- Networks
- Data centers
- Cloud services

Transactional Investments
Aim to reduce operational costs.
Ex:
- Payroll systems
- Inventory systems

Categories of IS Investments cont.

Informational Investments

Improve management decision-making.

Ex:

- Dashboards
- Business intelligence tools

Strategic Investments

Create competitive advantage.

Ex:

- AI systems
- E-commerce platforms

Strategic vs Operational Investments

Strategic Investments

Support long-term competitive advantage.

Ex:
- AI-driven customer service
- Digital transformation projects

Operational Investments
Improve day-to-day business activities.

Ex:
- Payroll systems
- Attendance management systems

Evaluating the IS Investments

Before investing, organizations must determine whether the investment is worthwhile.

Ex: Cost-Benefit Analysis (CBA)

Costs

- ✓ Hardware
- ✓ Software
- ✓ Training
- ✓ Maintenance
- ✓ Implementation

Benefits

- ✓ Increased productivity
- ✓ Cost savings
- ✓ Better customer satisfaction
- ✓ Revenue growth

Evaluating the IS Investments

Calculating the ROI and the Payback Time Period

Formula

$$ROI = \frac{Benefits - Costs}{Costs} \times 100$$

Measures profitability of an investment.

$$\text{Payback } Period = \frac{Initial\ Investment}{Annual\ Cash\ Inflow}$$

Example

- Initial Investment = Rs. 1,000,000
- Annual Cash Inflow (or Annual Savings) = Rs. 250,000
- Payback Period = 1,000,000 ÷ 250,000 = **4 Years**

- Shorter payback period → Lower risk and faster recovery of investment
- Longer payback period → Higher risk and slower return of investment

Risks in IS Investments

Technical Risks
Software failures
Integration issues

Financial Risks
Cost overruns
Unexpected expenses

Operational Risks
Business disruptions
Employee resistance

Security Risks
Cyberattacks
Data breaches

IT Governance ensures that information systems investments align with organizational goals with proper risk management and maximum benefits